

Experiment 2

Cross-Platform Prompting

Submission PDF

[Source folder on GitHub](#)

Prepared 2026-09-29 from the experiment materials on branch
``codex/complete-experiments``.

Source: [Ex02_Cross_Platform_Prompting/README.md](#)

Ex-2: Cross-Platform Prompting - Evaluating Diverse Prompt Engineering Techniques

Due: 10 Sep 2026 **Reference repo:** RaajaThilahar/Ex.No.2

Objective

Design prompts using different structures and show how one task improves from a basic prompt to a fully specified one.

Chosen application: Text Summarization

Source text: `source_text.md` (same input for every prompt level).

Prompt ladder

Level	File	What it adds
1 Basic	<code>prompts/1_basic.md</code>	Just the task
2 Role	<code>prompts/2_role.md</code>	Who the model is
3 Context	<code>prompts/3_context.md</code>	Audience and purpose
4 Constraint	<code>prompts/4_constraint.md</code>	Length, tone, exclusions
5 Output format	<code>prompts/5_output_format.md</code>	Exact structure

Procedure

Saved ChatGPT and Claude responses are in `outputs/level<N>_<tool>.md`; the provisional scoring is in `evaluation.md`. Response speed was not measured, and the ratings need independent human review.

Source: [Ex02_Cross_Platform_Prompting/outputs/README.md](#)

Save results as `level<N>_<tool>.md`, e.g. `level3_chatgpt.md`, `level5_claude.md`.

Source: [Ex02_Cross_Platform_Prompting/LAB_RECORD.md](#)

Lab Record: Ex-2: Cross-Platform Prompting

Field	Entry
Name / Reg. No.	_____
Date	_____
Repo	https://github.com/RHUDHRESH/PROMPT_ENGINEERING-

Aim

Design prompts with different structures (basic, role, context, constraint, output format) for text summarization.

Tools / Apparatus

Claude (saved earlier) and ChatGPT (current session)

Procedure

Prompts written (see prompts/ or the folder README).

Each prompt run on every tool; raw outputs saved under outputs/ (or the folder's equivalent).

Results scored in evaluation.md on: Faithfulness, constraint-following, format correctness, readability.

Findings summarised below.

Observations

See evaluation.md. Add your own scores for any tool you ran yourself. Tools not run are left blank; do not fill them in from memory.

Result

Five prompt levels have saved outputs for two tools and a provisional comparison. Context improved audience fit, constraints shortened the response, and the final format made key fields checkable.

Conclusion

Prompt structure can guide audience, length and output shape. Level 5 is the most verifiable, but concise constraints can reduce readability. Scores need independent review; speed was not measured.

Faculty signature: _____

Source: [Ex02_Cross_Platform_Prompting/evaluation.md](#)

Evaluation

Scores use 1-5. ChatGPT outputs were produced in the current Codex session; Claude outputs were already in the repository. The ChatGPT scores are provisional review by the model that produced those outputs, while the repository's Claude scores were originally self-assessed. Word counts and JSON can be checked mechanically; readability scores are subjective. No stopwatch was used.

Level	Tool	Faithful to source	Follows constraints	Format correct	Readability	Notes
1 Basic	ChatGPT	5	5	N/A	4	Covers the source's main claims without a requested length cap; concise.
1 Basic	Claude	4	5	N/A	4	Original repository score; covers the central points and names products.
2 Role	ChatGPT	5	5	N/A	5	Adds a headline and magazine-style paragraphs without unsupported facts.

Level	Tool	Faithful to source	Follows constraints	Format correct	Readability	Notes
2 Role	Claude	4	5	N/A	4	Original repository score; more magazine-like than level 1.
3 Context	ChatGPT	5	5	N/A	5	Explains token, hallucination and retrieval in plain language for the stated audience.
3 Context	Claude	4	5	N/A	5	Original repository score; explains several technical terms for students.
4 Constraint	ChatGPT	5	4	N/A	4	49-word reply and no company names. Some technical terms could be glossed more explicitly.
4 Constraint	Claude	4	4	N/A	3	Original repository score; 57 words, but dense and not all jargon is glossed.
5 Format	ChatGPT	5	5	5	4	JSON has the requested keys; headline is under 8 words, summary under 60 words, and limitations has two entries.
5 Format	Claude	4	5	5	3	Original repository score; JSON was reported as validated, with terse wording.

Observations

- Context and constraints change the content and audience fit most; formatting makes the output machine-checkable.
- Level 4 is 49 words in the ChatGPT output and uses no company names. Its explanations can still improve.
- Level 5 follows the requested structure. The fenced payload should be passed to `json.loads` if machine validation is required.
- The saved outputs are single samples; fresh identical runs and blind human scoring would make this comparison more reliable.

Final user-defined prompt

Level 5, with this added instruction: Write the summary in plain sentences rather than fragments, explain every technical term used, and include only facts stated in the source. It keeps audience context and produces verifiable fields while addressing the terse style seen in the saved samples.

Source: [Ex02_Cross_Platform_Prompting/outputs/level1_chatgpt.md](#)

Level 1 (Basic) - ChatGPT output

Transformers use self-attention to relate tokens in parallel, helping them learn long-range relationships faster than recurrent networks. Training on large text collections produced LLMs that predict the next token and are tuned to follow instructions. They can hallucinate, cost substantial compute and reflect biases. Research aims to improve efficiency, retrieval, context length and reliability.

Source: [Ex02_Cross_Platform_Prompting/outputs/level1_claude.md](#)

Level 1 (Basic) - Claude output

Transformers, introduced in 2017 in the paper "Attention Is All You Need", replaced recurrent networks as the leading architecture for sequence modelling. Their self-attention mechanism relates every token to every other token in parallel, which speeds up GPU training and captures long-range dependencies. Scaling them on large text corpora produced Large Language Models such as GPT, Claude and Gemini, which are pre-trained to predict the next token and then fine-tuned with human feedback to follow instructions. They can hallucinate, are costly to train and run, and inherit data biases. Research now targets efficiency, retrieval-augmented generation, longer context windows and more reliable reasoning.

Source: [Ex02_Cross_Platform_Prompting/outputs/level2_chatgpt.md](#)

Level 2 (Role) - ChatGPT output

Why transformers changed language technology

Transformers became the leading architecture for sequence modelling because self-attention lets a model compare tokens across an entire input at once. This helps capture long-range relationships and allows efficient parallel training. Large language models (LLMs) learn to predict the next token, then instruction tuning helps them respond to requests. Their output can still be wrong or biased, and training and use require substantial compute. Current work explores retrieval, efficiency and more reliable evaluation.

Source: [Ex02_Cross_Platform_Prompting/outputs/level2_claude.md](#)

Level 2 (Role) - Claude output

Transformers and the rise of LLMs

Since the 2017 paper "Attention Is All You Need", transformers have displaced recurrent networks as the standard architecture for sequence modelling. Rather than reading tokens one by one, they use self-attention to link every token with every other in parallel. This makes GPU training much faster and captures long-range dependencies.

Scaling transformers on large text corpora gave rise to Large Language Models such as GPT, Claude and Gemini. These are pre-trained to predict the next token, then fine-tuned with human feedback so they follow instructions.

The caveats: they can hallucinate facts, are expensive to train and run, and inherit biases from their training data. Active research areas are efficiency, retrieval-augmented generation, longer context windows and dependable reasoning.

Source: [Ex02_Cross_Platform_Prompting/outputs/level3_chatgpt.md](#)

Level 3 (Context) - ChatGPT output

Transformers process a sequence by letting each token weigh information from other tokens, rather than reading strictly one word at a time. This helps models connect distant ideas and makes training easier to parallelise. Large language models learn by predicting the next token in large collections of text, then are tuned with examples to follow instructions. They may still hallucinate (state false claims), inherit data bias, and consume significant computing resources. Retrieval-augmented generation supplies relevant documents when answering; longer context windows let a model consider more input, but do not guarantee accurate reasoning.

Source: [Ex02_Cross_Platform_Prompting/outputs/level3_claude.md](#)

Level 3 (Context) - Claude output

Sidebar: Why transformers changed language AI

Older language models, called recurrent networks, read text one word-piece (token) at a time. Transformers, introduced in 2017, instead use "self-attention": every token is compared with every other token at once. This parallel approach trains much faster on GPUs and links words that sit far apart in a text.

Training transformers on huge text collections produced Large Language Models (LLMs), such as GPT, Claude and Gemini. First they learn to predict the next token; then human feedback fine-tunes them to follow instructions.

They are not perfect. LLMs can hallucinate (state false things confidently), cost a lot to train and run, and absorb biases from their data. Researchers are working on efficiency, retrieval-augmented generation (looking up documents while answering), longer context windows and more reliable reasoning.

Source: [Ex02_Cross_Platform_Prompting/outputs/level4_chatgpt.md](#)

Level 4 (Constraint) - ChatGPT output

Transformers compare text pieces at once, which helps connect distant ideas and speeds training. Large language models (LLMs, text-prediction systems) learn by predicting the next piece, then use examples to follow instructions. They can invent facts, repeat data bias and cost much to run. Document lookup adds useful sources when answering. Longer input limits do not guarantee correct answers.

Word count: 49. No company names. Text pieces is a short gloss for tokens; invent facts explains hallucination without adding jargon.

Source: [Ex02_Cross_Platform_Prompting/outputs/level4_claude.md](#)

Level 4 (Constraint) - Claude output

Transformers (attention-based neural networks), introduced in 2017, replaced recurrent networks (step-by-step sequence readers). Self-attention (comparing all tokens) works in parallel, speeding training and capturing long-range links. Scaled on large text and fine-tuned with human feedback, they became large language models. They can hallucinate, are costly, and inherit data biases; research targets efficiency, retrieval, longer context, reliable reasoning.

Source: [Ex02_Cross_Platform_Prompting/outputs/level5_chatgpt.md](#)

Level 5 (Output format) - ChatGPT output

```
{
  "headline": "How language models use transformers",
  "summary": "Transformers compare text pieces in parallel, helping models connect distant ideas. Large language models learn by predicting the next piece and then use examples to follow instructions. They can invent facts, reflect data bias and require substantial computing resources. Retrieval supplies relevant documents at answer time.",
  "key_terms": [
    {"term": "token", "meaning": "text piece"},
    {"term": "retrieval", "meaning": "document lookup"}
  ],
  "limitations": [
    "Generated claims may be false.",
    "Training and use require substantial compute."
  ]
}
```

Source: [Ex02_Cross_Platform_Prompting/outputs/level5_claude.md](#)

```
{ "headline": "Transformers Replaced Recurrent Networks in Language AI", "summary": "Transformers, introduced in 2017, use self-attention to compare all tokens in parallel, training faster than recurrent networks and capturing long-range links. Scaled on large text and fine-tuned with human feedback, they became large language models. They can hallucinate, are costly, and inherit biases; research targets efficiency, retrieval, longer context and reliable reasoning.", "key_terms": [ {"term": "Self-attention", "meaning": "Token-to-token comparison"}, {"term": "Recurrent network", "meaning": "Step-by-step sequence reader"}, {"term": "Token", "meaning": "Small text piece"}, {"term": "Hallucination", "meaning": "Confident false output"}, {"term": "Retrieval-augmented generation", "meaning": "Looks up documents"} ], "limitations": [ "Can hallucinate facts and are expensive to train and run.", "Inherit biases from their training data." ] }
```

Source: [Ex02_Cross_Platform_Prompting/prompts/1_basic.md](#)

Level 1 - Basic

Summarize this text.

<paste source_text.md>

Source: [Ex02_Cross_Platform_Prompting/prompts/2_role.md](#)

Level 2 - Role

You are a technical editor for an engineering magazine. Summarize this text.

<paste source_text.md>

Source: Ex02_Cross_Platform_Prompting/prompts/3_context.md

Level 3 - Context

You are a technical editor for an engineering magazine.
Context: the readers are final-year students who know basic ML but have not studied transformers. The summary will appear as a sidebar next to a longer article on LLMs.
Summarize this text.

<paste source_text.md>

Source: Ex02_Cross_Platform_Prompting/prompts/4_constraint.md

Level 4 - Constraint

You are a technical editor for an engineering magazine.
Context: the readers are final-year students who know basic ML but have not studied transformers. The summary will appear as a sidebar next to a longer article on LLMs.
Constraints: max 60 words; neutral tone; no jargon without a 3-word explanation; do not mention company names; do not add facts that are not in the text.
Summarize this text.

<paste source_text.md>

Source: Ex02_Cross_Platform_Prompting/prompts/5_output_format.md

Level 5 - Output format

You are a technical editor for an engineering magazine.
Context: the readers are final-year students who know basic ML but have not studied transformers. The summary will appear as a sidebar next to a longer article on LLMs.
Constraints: max 60 words in the summary; neutral tone; explain jargon in <=3 words; no company names; no facts outside the text.

Output format (JSON only):

```
{
  "headline": "<=8 words",
  "summary": "<=60 words",
  "key_terms": [{"term": "", "meaning": ""}],
  "limitations": ["", ""]
}
```

Summarize this text.

<paste source_text.md>

Source: Ex02_Cross_Platform_Prompting/source_text.md

Transformers, introduced in 2017 in "Attention Is All You Need", replaced recurrent networks as the dominant sequence-modelling architecture. Instead of processing tokens one at a time, they use self-attention to relate every token to every other token in parallel, which makes training far faster on GPUs and captures long-range dependencies. Scaling transformers on large text corpora produced Large Language Models such as GPT, Claude and Gemini. These models are pre-trained to predict the next token, then fine-tuned with human feedback so they follow instructions. Despite strong results, they can hallucinate facts, are expensive to train and run, and inherit biases from their data. Current research focuses on efficiency, retrieval-augmented generation, longer context windows and reliable reasoning.